Overview of Stellar Cyber

00:00-01:00
Welcome to the Overview of Stellar Cyber. Start your journey to a human-augmented autonomous SOC. One SecOps platform that includes AI-driven SIEM, NDR, OT, ITDR, UEBA, TIP, Open XDR, and multilayer AI. Here is the Stellar Cyber platform user interface. You will learn more about the operations and capabilities this Stellar interface provides throughout this overview. In this overview you will learn about the core operations and key capabilities of the Stellar Cyber platform including data collection and normalization, data enrichment, Interflow records, storage--including the Data Lake and Data Sink to cold storage, threat detection, correlation, and alerting;

01:00-02:00
and finally, response actions with playbooks. Let us get started, along the way you will see the progress. Let us start with data collection and normalization. To power AI-driven threat detection and automated response, Stellar Cyber collects and normalizes data from on-premises and cloud sources. Let us begin by looking at how data is collected and normalized from on-premises infrastructure. As an open and unified platform, Stellar Cyber includes hundreds of integrations to collect and normalize data from on-premises infrastructure, security, and identity tools and systems. Here, on the right of the diagram, is the Stellar Cyber platform that can be a SaaS instance, a cloud-native, cloud-hosted deployment, or an on-premises deployment.

02:00-03:00
On-premises or private cloud-hosted tools forward logs or syslogs and/or network traffic to Stellar Cyber Modular Sensors. Here are some common infrastructure and security tools including firewalls; network devices, such as routers and switches; endpoint detection and response solutions; vulnerability management tools; and network traffic from SPAN ports or TAP devices. A key component to the Stellar Cyber architecture are Modular Sensors. They are deployed within the network as virtual or hardware appliances. Security and network tools are configured to send logs to Stellar Cyber Modular Sensors that are log forwarders for the Stellar Cyber platform. Modular Sensors can collect network traffic from switches using SPAN or mirror ports or TAP devices.

03:00-04:00
Stellar Cyber Modular Sensors include network detection and response (NDR) capabilities. Along with collecting logs, they collect metadata from network traffic, check IDS signatures for IT and OT threats, check for malware, and normalize data into JSON Interflow records, which are securely sent to the Stellar Cyber platform over a TLS connection. Next,

connectors that run on Stellar Cyber Modular Sensors can collect security data from on-premises or private-hosted security tools using APIs or LDAPS. Here Microsoft Active Directory servers are deployed on premises as well as vulnerability management tools such as Tenable.

04:00-05:00
Stellar Cyber Modular Sensors have built-in connectors for integration with third-party security and identity systems. With the Modular Sensor being within the organization, they can securely connect to Active Directory. They are able to obtain important user information to enrich Interflow records with valuable context. The connectors to Active Directory can also be used to block user accounts that have been compromised. Connectors to vulnerability tools such as Tenable can obtain valuable scan data as well as launch scans within the organization. After collecting the information, the Modular Sensor normalizes it into JSON-formatted Interflow records. These Interflow records are securely transmitted to the Stellar Cyber platform over an encrypted TLS connection.

05:00-06:00
Stellar Cyber server sensors collect logs and event data from Windows servers and Linux servers. The server sensors run on Windows servers and Linux servers as shown here. They normalize collected logs and event data into JSON Interflow records and securely sends them to the Stellar Cyber platform over a secure TLS connection. Building on how Stellar Cyber collects and normalizes data for AI-driven threat detection and response, we will now look at how this applies to systems and services operating within cloud infrastructure. The Stellar Cyber platform also includes a comprehensive set of integrations to collect and normalize data from cloud applications,

06:00-07:00
cloud infrastructure, identity and access management systems, and cloud security tools and services. The Stellar Cyber platform has built-in connectors specifically for cloud-accessible services. Cloud security tools and services that can be accessed via connectors via API include: EDR (Endpoint Detection and Response) solutions, email systems, SASE (Secure Access Service Edge) solutions, IAM (Identity and Access Management) providers, Cloud apps, Cloud infrastructure, and Cloud security services. Stellar Cyber platform Connectors collect raw events and security data from cloud applications, cloud infrastructure, identity and access management providers, and a variety of cloud security tools and services.

07:00-08:00
In this example, the Stellar Cyber platform Connectors use APIs to request and collect raw events and security data directly from cloud security tools and services. After the response is received, the Stellar Cyber platform normalizes the collected raw events and security data into JSON Interflow records for secure processing. Next, I will explain Stellar Cyber data

enrichment. Metadata collected via connectors, sensors, and integrated threat-intelligence feeds is used to enrich Interflow records with valuable context. Now here is an example of how the enrichment process works. The Stellar Cyber platform has an Interflow record received from a log source sensor or connector as shown here. The Stellar Cyber platform enriches the data within Interflow records from many sources including threat-intel feeds for IP and URL reputation,

08:00-09:00
geolocation, DNS for IP-to-hostname mapping, and identity management solutions including Microsoft Entra ID, formerly known as Azure AD; Active Directory; Okta; Google Workspace; etcetera. The Stellar Cyber platform enriches Interflow records collected from all data sources with contextual metadata derived from threat intelligence feeds -- IP and URL reputation sources, malicious IPs, domains, file hashes, and indicators of compromise (IOCs). Let us take a closer look at the threat intel platform built into Stellar Cyber. Here is an example of the threat intel platform settings in Stellar Cyber. It comes with preconfigured threat intelligence feeds such as AlienVault, DHS, ET Pro, PhishTank, Abuse.ch, OpenPhish, and Stellar Cyber.

09:00-10:00
You can bring in premium feeds such as recorded future, SOC Radar, and CrowdStrike or use any standard TAXII feed from your trusted sources. The threat intel platform will collect threat intel at ingestion. It is not only used for enrichment. This way, every IOC adds context for raw data so you catch threats earlier. Now let us continue with enrichment. Interflow records are also enriched with geolocation data: country, city, ASN, and location-based risk indicators. Interflow records are also enriched with user metadata from identity providers: usernames, user details, privileges, roles, group memberships, and login activity. Enrichment also includes asset metadata: host names, OS, risk scores, location, vendors, types, file access commands, logins, login failures, alerts, and vulnerabilities.

10:00-11:00
Interflow records are also enriched with the XDR Kill Chain and MITRE ATT&CK framework tactic and technique classification. Modular Sensors also enrich Interflow records with application and protocol identification, deep packet inspection (DPI) for over 4,000 applications and protocols including DNS, SMB, HTTP, ICMP, Google Drive, Dropbox, PowerShell, RDP, Windows management instruction, etcetera. Here is an example showing the Stellar Cyber platform getting metadata from an enrichment service, which is then used to enrich an Interflow record. The enrichment performed by Stellar Cyber is extremely powerful adding valuable context to identify and investigate threats.

11:00-12:00
Now it is time to learn about Stellar Cyber Interflow records. Stellar Cyber Interflow records. Data that is collected and enriched by Stellar Cyber from all sources is normalized into standard JSON-formatted records called Interflow records. Interflow records are stored in Stellar Cyber and are optimized for threat detection and investigation using AI and machine learning. Here is an example of the data sources Stellar Cyber uses to collect important data used in creating Interflow records. Here is an example of an Interflow record in the Stellar Cyber platform. It has the normalized data as well as the data added during enrichment.

12:00-13:00
Here you can see the Interflow record for this alert has been enriched with source and destination geolocation metadata, threat intel: IP and URL reputation metadata, source and destination IP type metadata, and IDS engine telemetry from Modular Sensors. Interflow records are also enriched with user data, asset data, Kill Chain data, IOC matches, DNS domains, application names, file hashes, and much more depending on the type of alert. storageNow let us learn about storage in Stellar Cyber with the Data Lake and Data Sink. Storage: retention of Interflow records within the Stellar Cyber platform. The Stellar Cyber platform includes a Data Lake.

13:00-14:00
Stellar Cyber also provides cold storage options as a cost-effective way to store data for a longer term. Enriched Interflow records are stored in the Data Lake within the Stellar Cyber platform and can be simultaneously replicated to long-term storage using Stellar Cyber Data Sink. For on-premises deployments, platform admins can use Data Sink with Amazon S3 or compatible cold storage systems. For SaaS deployments Stellar Cyber manages the cold storage. The Stellar Cyber Ingestion Dashboards show data stored within the Stellar Cyber platform from each source. Traffic records over time is data stored from network detection and response (NDR) sources,

14:00-15:00
such as Modular Sensors collecting traffic from the network. Traffic by application is available with the included NDR capabilities that collect metadata from traffic and performs deep packet inspection to determine the applications. Traffic by top sources shows the amount of data collected from each source, based on the time selected. You can view built-in or create custom dashboards as seen here to view data collected by the Stellar Cyber platform. Now let us learn about threat detection, correlation, and alerting. The Stellar Cyber platform uses multi-layer AI for detection, correlation, and triage. Enriched Interflow records in the Data Lake are ready to be analyzed.

15:00-16:00
Stellar Cyber's multilayer AI consists of machine learning (ML) for threat detection and anomaly detection. The Interflow records in the Stellar Cyber Data Lake are optimized for multilayer AI with machine learning. Machine learning is used to detect threats by analyzing patterns and data to identify anomalies, behaviors, or indicators that match known or emerging attack techniques. Graph machine learning is used for alert correlation and case creation. Graph machine learning is used to correlate and link related alerts, users, assets, behaviors, and activities into cases, delivering a unified view that simplifies triage and accelerates Investigations.

16:00-17:00
Gen AI provides capabilities such as an automated triage agent, AI investigator, and case summarization. Gen AI is used for verdict determination for auto triage, case analysis, case summaries, and for using plain language questions to quickly uncover key insights across your data. The Stellar Cyber platform also includes rules, logic, and playbooks, including signature-based rules, heuristic logic, and custom ATH playbooks. AI is complemented by signature-based rules, heuristic logic, and playbooks to deliver broader threat coverage, more precise detections, and automated response. The Stellar Cyber platform user interface organizes the data processed by multi-layer AI

17:00-18:00
along with rules and logic so that alerts and events can be quickly found, investigated, and actioned. Alerts detected by the Stellar Cyber platform are automatically categorized into the MITRE ATT&CK framework and the XDR Kill Chain. The XDR Kill Chain enhances MITRE ATT&CK-based tactics and techniques with its own tactics to cover threat intelligence, malware detection, and anomaly detections. Here you can see the alerts that Stellar Cyber has categorized into the MITRE ATT&CK framework. Here the XDR Kill Chain shows how attacks progress across different stages making it easy to see where an attacker is in their campaign. It begins with initial attempts. These are the alerts detected where attackers probe for weaknesses and attempt to gain access.

18:00-19:00
Next are alerts mapped to persistent footholds where they establish ongoing access and work to avoid detection. Then comes alerts mapped to exploration where attackers search the environment, discover assets, and collect data. The next stage shows alerts mapped to propagation. This is where attackers spread across systems through credential use, privilege escalation, and lateral movement. The final and most critical stage shows the alerts mapped to exfiltration and impact where sensitive data is stolen or systems are disrupted. To see multi-layer AI in action for detection, correlation, and triage, here is an example of a case. Cases are automatically created from alerts, users, assets, processes, and activities,

collected from different data sources that are correlated automatically using graph machine learning.

19:00-20:00
Gen AI is used to create a case summary that explains what happened, why it matters, and what to do now. Gen AI is also used to provide the hypothesis for the case, which summarizes the situation, articulating the likely cause, affected assets, and business risk to guide immediate response. Graph machine learning is used to correlate alerts, users, assets, processes, activities, and other observables, into a case for holistic, automated threat detection and rapid incident response. Gen AI provides recommended steps for investigation,

20:00-21:00
vulnerability assessments, prioritizing monitoring efforts as well as actions for engaging incident response. Case actions can be performed to mitigate threats using third-party security integrations or scripts using case-specific attributes such as usernames, host names or IP addresses. Now let's learn about response actions with playbooks. Here I will show you response actions using on-premises or private cloud-hosted security tools. Stellar Cyber's platform empowers security teams to automate playbooks instantly taking containment actions the moment a threat is detected. Its true strength lies in flexibility, enabling custom threat-hunting searches to become automated triggers that transform discoveries into permanent defenses.

21:00-22:00
Here is how response actions work with on-premises security tools. In this example, in the company network there are on-premises or private cloud-hosted security and identity tools such as: Active Directory, other identity providers (IdPs), firewalls, and vulnerability management tools. On the right there is a Stellar Cyber platform instance that is available as SaaS, private cloud, or on premises. Within the company network, there are also one or more Stellar Cyber Modular Sensors that are hardware or virtual appliances. From inside the company network, Modular Sensors open a secure configuration management (CM) connection to the Stellar Cyber platform. Modular Sensors have built-in connectors that can communicate locally,

22:00-23:00
in this case with systems and services within the company network. Next, let us look at an automated threat hunting playbook example configured in Stellar Cyber that will use connectors on Modular Sensors to perform the actions on systems or services within the company environment. Here is an automated threat hunting playbook example. The configuration starts by defining the execution frequency of the automated response. This lets you specify how often you want this playbook to run, which can be in minutes, hours,

days, or weeks. Here you define the content to be monitored. A query or correlation rule determines which activities or data patterns the playbook will watch for helping the system automatically detect and respond to threats that match your custom logic.

23:00-24:00
Here you define the condition that triggers the response action. Defining these conditions allows the playbook to recognize specific threat patterns and automatically initiate the right response. Here is where you define the response action. ATH playbooks in Stellar Cyber can take a range of automated actions from disabling accounts and blocking IPs to containing hosts through EDR platforms, launching investigations, and notifying teams. These responses transform threat detections into immediate, coordinated defense. Now that the playbook is configured, let's take a look at it in action. Here are some examples of response actions: disable a user in Active Directory, block an IP address in a firewall, send an email or Slack notification, run a webhook or script.

24:00-25:00
After a playbook is configured and scheduled, the Stellar Cyber platform executes it automatically. When a trigger condition is met, the platform routes the response action through the CM connection to the Modular Sensor connector that interfaces with the on-premises service, such as Active Directory, to disable a user or a firewall to block an IP address. ATH playbooks can also automate response actions using cloud security tools and services. The Stellar Cyber Open XDR platform includes hundreds of built-in integrations across cloud applications, infrastructure, identity, and security tools, enabling both automated and manually triggered playbooks to mitigate threats.

25:00-26:00
As an Open XDR platform, Stellar Cyber includes pre-built integrations with a wide range of cloud security tools and services including: EDR (endpoint detection and response) solutions, email systems, cloud apps, cloud infrastructure, cloud security services, SASE (secure access service edge) solutions, and IDP (identity provider) solutions to name a few. Connectors used by playbooks to access cloud services. Example response actions include: disabling a user in an identity provider, performing a remediation action in an EDR solution, blocking an IP address in a firewall, running a script, running a webhook to open a ticket and sending email and Slack notifications to alert your security team.

26:00-27:00
When an ATH playbook triggers a cloud-based action, the Stellar Cyber platform uses its connector to deliver the response action to the appropriate cloud security or identity service for execution. ATH playbooks in Stellar Cyber deliver powerful automated response actions across both on-premises and cloud-security systems. With the ability to seamlessly integrate with hundreds of existing tools and services, they enable faster containment,

smarter remediation, and continuous protection. Moving security operations closer to a truly human-augmented, autonomous SOC.

27:00-28:00
We have completed the overview of the Stellar Cyber platform from data collection and normalization through enrichment and Interflow records to storage in the Data Lake and Data Sink. We then explored advanced threat detection, correlation and alerting, and finished with automated response actions using playbooks. Together, these capabilities deliver a unified, intelligent, and automated security operations platform. Thank you for watching the Overview of the Stellar Cyber SecOps platform that includes AI-driven SIEM, NDR, OT, ITDR, UEBA, TIP, Open XDR, and multi-layer AI. Start your journey to a human-augmented autonomous SOC. Thank you and happy learning!